CareerCompass: An Integrated Resume-Intelligent Job Matching and Placement Management Platform for College Students

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***Abstract*—The transition from academic education to employment requires students to manage resumes, discover suitable opportunities, understand skill requirements, submit applications, and track recruitment outcomes. Conventional placement and job-search systems often separate these activities, creating a fragmented workflow in which students must interpret job requirements and evaluate their own suitability manually. This paper presents CareerCompass, an integrated web-based career guidance and placement management platform that combines role-based placement workflows with resume intelligence and job-oriented career support. The implemented system provides Student, Recruiter, and Administrator modules, together with resume upload and replacement, structured resume analysis, OCR fallback for difficult PDF documents, Resume AI scoring, Resume Builder, hybrid Job Match, Skill Gap analysis, course recommendations, saved jobs, application tracking, notifications, recruiter feedback, company verification, and administrative management. The platform uses Flask and JavaScript-based web interfaces, PostgreSQL as the primary relational database, and Supabase for cloud database services and private resume storage. Job Match combines deterministic requirement evaluation with local semantic similarity and a safe TF-IDF fallback. Security hardening includes session-bound CSRF protection, role and ownership authorization, secure session-cookie configuration, PDF validation, and private resume storage. The final automated validation recorded 287 passed tests, 5 intentionally skipped tests, and no failures. This work demonstrates an integrated approach in which resume understanding, job matching, skill-gap identification, learning support, and placement workflows are connected within a single platform.**

***Index Terms*—career guidance, placement management, resume intelligence, job matching, skill gap analysis, ATS evaluation, recruitment management, PostgreSQL, Supabase, web-based platform**

I. Introduction

The transition from academic education to professional employment requires students to perform several interconnected activities, including career planning, profile development, resume preparation, job and internship searching, application management, and recruitment preparation. In educational institutions, these activities are frequently supported through separate platforms or manual processes. This fragmentation can make it difficult for students to maintain a consistent career profile and monitor their progress throughout the recruitment process. At the same time, recruiters and placement administrators require organized mechanisms for managing job opportunities, applications, candidates, and placement activities.

Existing placement management systems primarily focus on administrative and recruitment-related operations such as student registration, job posting, application processing, and placement records. Research has also investigated specialized solutions for skill-gap analysis, career recommendation, resume analysis, personalized career guidance, and Applicant Tracking System (ATS) screening. However, these capabilities are generally addressed as individual solutions rather than as components of a unified career-development and placement workflow. The literature reviewed for this study indicates a need for greater integration between student career preparation and institutional recruitment management.

To address this problem, this paper presents CareerCompass, an integrated web-based career guidance and placement management platform for college students. The system is designed around three primary user roles: students, recruiters, and administrators. Students can manage their profiles, explore job and internship opportunities, submit applications, and monitor recruitment progress. Recruiters can manage company information, publish opportunities, review applications, and manage candidate recruitment activities. Administrators can manage users, recruitment-related information, and overall placement operations.

The primary objective of CareerCompass is to provide a centralized platform that connects career preparation and placement activities within a common system. The platform is designed to reduce fragmentation between different stages of the recruitment workflow while providing role-specific functionality for students, recruiters, and administrators. The

system is implemented as a web application with PostgreSQL and Supabase services supporting interaction and persistent storage among the different user roles.

The current study focuses on the implemented CareerCompass platform and its evaluation through automated functional and performance-oriented testing. The implemented intelligent components include structured resume analysis, OCR fallback, Resume AI scoring, hybrid Job Match, Skill Gap analysis, and course recommendations.

The main contributions of this work are as follows:

- Development of an integrated platform combining career-development and placement-management workflows.
- Design of role-based functionality for students, recruiters, and administrators.
- A structured resume-processing pipeline supporting PDF validation, text extraction, OCR fallback, resume replacement, and Resume AI evaluation.
- A hybrid Job Match mechanism combining skills, experience, education, project relevance, and semantic similarity.
- An integrated Skill Gap and Courses workflow connecting job requirements with actionable learning support.
- Establishment of a PostgreSQL and Supabase-supported architecture with private resume storage.
- Security and reliability hardening validated through an expanded automated regression suite.

II. Related Work

Research on career development and placement management has addressed several individual aspects of the transition from education to employment. The reviewed literature includes studies on skill-gap analysis, career recommendation, resume screening, personalized career guidance, Applicant Tracking System (ATS) evaluation, and placement management. These studies provide the foundation for identifying the research gap addressed by CareerCompass.

A. Skill Gap Analysis

Li et al. investigated skill demand prediction and workforce skill-gap analysis within the manufacturing industry [1]. Their work focused on identifying differences between the skills required by employers and the skills available within the workforce. The study demonstrates the importance of analyzing industry skill requirements when identifying gaps between education and employment. However, the work is focused on skill-gap identification and does not provide an integrated platform for personalized career guidance, resume management, job matching, or placement activities.

B. AI-Based Career Guidance

Research has also explored artificial intelligence and machine learning for personalized career recommendation [2], [4]. These studies investigated AI-powered career recommendation using psychometric information, machine learning techniques, and established career-development theories to identify suitable career paths for individuals. These studies demonstrate the potential of intelligent recommendation systems for supporting career decisions. However, their primary focus is career recommendation and guidance rather than integrating these functions with resume preparation, job applications, recruiter interaction, and placement management.

C. Resume Analysis and ATS Screening

AI-based resume analysis has been investigated to automate resume screening and provide recommendations to candidates or recruiters [3]. Existing approaches use natural language processing and automated analysis to extract information from resumes and evaluate their suitability. Research on fair and ethical ATS screening has further examined the use of natural language processing and generative AI for resume evaluation while considering potential fairness and bias concerns [5]. These studies demonstrate the value of intelligent resume assessment, but they primarily address resume screening and evaluation rather than the complete career-development and placement workflow.

D. Placement Management Systems

Traditional placement management research has focused on automating activities involving students, companies, and administrators. Such systems provide functions including student registration, company management, job posting, application processing, and placement record management [6]. Review studies of training and placement management systems similarly emphasize the importance of centralized management and automation of recruitment-related activities [7]. However, these systems generally focus on administrative placement operations and do not provide an integrated layer of personalized career development and intelligent career support.

E. Comparative Analysis

The reviewed literature demonstrates that existing systems address important but generally specialized aspects of career development and recruitment. Skill-gap analysis focuses on identifying differences between workforce capabilities and industry requirements, while career recommendation systems focus on identifying suitable career paths. Resume-analysis and ATS systems concentrate on evaluating candidate documents, whereas placement-management systems primarily automate recruitment and administrative activities.

This separation indicates an opportunity to integrate career preparation and placement workflows within a common platform. CareerCompass is designed to address this integration by connecting student career activities, job and internship opportunities, application management, recruiter workflows, and administrative placement operations within a unified system. The intelligent capabilities described in the broader CareerCompass design, including resume analysis, ATS evaluation, skill-gap analysis, career recommendation, and course recommendation, are planned as future extensions of the platform rather than being treated as implemented components in the current study.

III. Research Gap and Problem Formulation

The literature reviewed in this study demonstrates that career development and placement management have been investigated from several different perspectives. Existing studies provide solutions for individual problems such as workforce skill-gap analysis, personalized career recommendation, AI-based resume screening, ATS evaluation, and placement management. However, these solutions generally concentrate on a specific stage of the career-development or recruitment process rather than integrating the complete workflow within a common platform.

The reviewed skill-gap analysis research primarily focuses on identifying differences between required and available skills. Career recommendation studies concentrate on recommending suitable career paths based on user characteristics, psychometric information, or career-development theories. Resume-analysis and ATS research focuses on evaluating candidate resumes and improving automated screening. In contrast, placement-management systems mainly address administrative activities involving students, companies, applications, and placement records.

The identified gap is therefore the limited integration between career-development support and placement-management activities within a unified student-centered platform. In particular, the reviewed literature does not demonstrate a single integrated workflow that combines structured resume understanding, job-specific matching, skill-gap identification, learning support, application tracking, recruiter interaction, and administrative placement management.

CareerCompass addresses this gap by providing a centralized platform connecting students, recruiters, and administrators. The implemented system integrates profile management, job management, application workflows, recruiter-side candidate management, notifications, resume intelligence, Job Match, Skill Gap analysis, Courses, and administrative placement operations.

A. Problem Formulation

The research problem addressed by this study is the fragmentation of career-development and campus-recruitment activities across separate workflows and systems. Students may need to manage their career information, search for opportunities, submit applications, and monitor recruitment progress through disconnected processes, while recruiters and administrators require separate mechanisms for managing opportunities, candidates, and placement activities.

Therefore, this study investigates the following problem:

> How can career-development and campus-recruitment activities be integrated into a unified web-based platform that supports students, recruiters, and administrators while providing measurable functional, usability, and performance characteristics?

B. Research Objectives

The objectives of CareerCompass are:

- To develop a centralized platform for managing student career and placement activities.
- To integrate student, recruiter, and administrator workflows within a common system.
- To analyze uploaded resumes and derive structured career-related information.
- To evaluate resumes using ATS-oriented and content-quality metrics.
- To match students with jobs using requirement-based and semantic signals.
- To identify missing skills and connect them with learning resources.
- To provide recruiters with facilities for managing opportunities, applicants, statuses, and feedback.
- To provide administrators with centralized management of users, recruiters, jobs, and placement-related information.
- To evaluate the implemented platform through automated functional and performance-oriented testing.

C. Research Questions

The study is guided by the following research questions:

- **RQ1:** How effectively can resume intelligence and placement workflows be integrated into a unified platform?
- **RQ2:** How can multiple resume-job compatibility signals be combined to provide explainable job matching?
- **RQ3:** What functional and performance characteristics does the implemented CareerCompass platform demonstrate?

IV. CareerCompass System

CareerCompass is an implemented centralized web-based platform that connects the major participants in the college recruitment process. The system is organized around three primary roles: Student, Recruiter, and Administrator. Each role is provided with specific functionality while sharing common application, database, and cloud services.

A. Student Module

The Student Module provides students with facilities for managing their career-related information and participating in recruitment activities. The module includes registration and authentication, profile management, dashboard access, job discovery, search and filtering, job details, saved jobs, application submission and tracking, notifications, resume upload and replacement, Resume AI, Resume Builder, Job Match, Skill Gap, Courses, and account settings.

The student workflow connects multiple activities within a single platform. A student can maintain profile information, explore available opportunities, submit applications, upload or build a resume, and monitor recruitment progress without moving between independent systems.

B. Resume Intelligence

The resume pipeline begins with validated PDF upload. Selectable PDF text is extracted locally, and OCR is used as a fallback when normal extraction is insufficient. The extracted

content is converted into structured information such as skills, education, experience, projects, certifications, and missing fields. The structured analysis is passed to the canonical Resume AI scoring provider so that Resume and Resume AI interfaces use the same underlying metrics.

C. Resume Builder

The Resume Builder provides multiple templates, including Classic, Modern, Minimal, and Creative. The builder allows students to generate structured resumes while maintaining consistency with the resume-management workflow. The Creative template is supported both for saving the selected template and for PDF generation.

D. Job Match

Job Match ranks active jobs according to the student's active resume. The current hybrid approach combines deterministic components with semantic similarity:

$$M = 0.45S + 0.15E + 0.10D + 0.10P + 0.20Q \quad (1)$$

where S is required-skills match, E is experience fit, D is education fit, P is project relevance, and Q is semantic similarity.

The semantic component uses a local Sentence Transformer model based on *all-MiniLM-L6-v2*, with a TF-IDF/scikit-learn fallback when the semantic model is unavailable. The implementation reuses a singleton model and computes the resume embedding once per matching operation. Skill normalization also protects against common false-positive matches such as Java versus JavaScript and SQL versus MySQL or PostgreSQL.

The Jobs module uses database-level pagination for browser requests. Job Match evaluates a bounded candidate set of the newest active jobs to control computational cost as the number of active jobs grows.

E. Skill Gap and Courses

Skill Gap analysis connects job requirements with the skills extracted from the student's active resume. Missing or insufficiently represented skills are surfaced to the student. The Courses module provides learning-oriented recommendations associated with identified career or skill requirements. Together, these modules create a feedback loop from job requirements to learning and subsequent resume improvement.

F. Recruiter Module

The Recruiter Module provides recruiters with functionality for managing recruitment activities. Recruiters can maintain company-related information, manage company profiles and verification-related workflows, publish opportunities, manage posted jobs, view applicants, review resumes, update application statuses, and provide recruiter feedback or rejection reasons.

The module provides a recruiter-side workflow corresponding to the student application process. Applications submitted by students can be accessed by recruiters for further recruitment activities, including application-status updates.

G. Administrator Module

The Administrator Module provides centralized management of the platform. Administrative functions include management of student accounts, recruiter accounts, company registration requests, job-related information, and other placement-related activities.

The administrator role provides an additional layer of control over the overall platform and supports centralized monitoring and management of information used by students and recruiters.

H. Application and Recruitment Workflow

CareerCompass connects student applications with recruiter-side candidate management. The general workflow begins when a recruiter registers a company and receives administrative approval. The recruiter can then publish an opportunity. Students can view the opportunity and submit an application with an appropriate resume. The application can then be accessed by the recruiter for candidate management and status updates, while students can monitor the resulting status through their application tracker and notifications.

The integrated workflow can be represented conceptually as:

Company/Recruiter → Job/Internship → Student Application → Recruiter Review → Recruitment Status

The detailed implementation and status transitions are evaluated based on the functionality available in the implemented system.

V. System Architecture and Implementation

CareerCompass is implemented as a web-based application using a client-server architecture. The current project structure contains a Flask application, a database access layer, HTML templates, CSS stylesheets, JavaScript files, PostgreSQL database services, and Supabase cloud services. The application provides separate interfaces for student, recruiter, and administrator roles.

A. Technology Stack

The current implementation uses Python with the Flask web framework for backend application development. HTML templates construct the server-rendered user interface, while CSS and JavaScript provide presentation and client-side functionality. PostgreSQL is the authoritative relational database, while Supabase provides cloud database connectivity and private object storage.

The main project components include the Flask application, database access module, HTML templates, CSS resources, JavaScript resources, PostgreSQL database, and Supabase services.

B. Backend Architecture

The backend follows a request-processing workflow in which client requests are received by the Flask application and processed through the appropriate application logic and database operations.

The general architecture is:

Client → Flask Application
→ Database Layer → PostgreSQL / Supabase

The implemented routes, authentication mechanisms, database operations, role-based authorization logic, and application workflows are evaluated during system testing.

C. Database Management

A centralized PostgreSQL database is used to persist application information. The database supports interaction between the Student, Recruiter, and Administrator modules and provides persistent storage for the entities required by the implemented workflows.

Supabase is used as the cloud service layer supporting database connectivity and private storage functionality. Resume files are stored using private Supabase Storage, while associated resume metadata and application information are maintained within the PostgreSQL database.

The database structure and relationships are represented through the implemented relational schema and are validated during database and workflow testing.

D. Resume Storage and Validation

Resume management is integrated into the application workflow. When a student uploads a resume, the Flask backend validates the uploaded file before storing it in private Supabase Storage. Resume metadata is associated with the corresponding student and application records in PostgreSQL.

The implemented workflow can be represented as:

Student → PDF Validation → Supabase Storage
→ PostgreSQL Metadata → Authorized Recruiter Access

This architecture separates resume-file storage from application metadata while maintaining controlled access through the application.

E. Role-Based Access

CareerCompass provides different interfaces and workflows according to the user role. Students, recruiters, and administrators are provided with separate authentication and dashboard functionality. Access to role-specific operations is controlled by the application logic.

The role-based structure is intended to prevent users from accessing functions that belong to another user category while maintaining a common platform.

F. Security Architecture

Security hardening includes session-bound CSRF protection for unsafe requests, role-based authorization, resource-ownership checks, secure session-cookie configuration, production debug mode disabled by default, server-side PDF validation, file-size limits, private resume storage, and environment-based configuration for sensitive secrets. These controls are applied across Student, Recruiter, and Administrator mutations.

VI. Methodology and Experimental Design

The study follows a development and evaluation process consisting of requirement analysis, literature review, system design, implementation, integration, testing, and evaluation. The methodology is designed to evaluate the implemented CareerCompass platform without assuming functionality that has not yet been developed.

A. Development Methodology

The system development process follows the following stages:

1) Requirement analysis based on the CareerCompass Software Requirements Specification.
2) Review of existing research related to career guidance and placement management.
3) Design of the multi-role CareerCompass architecture.
4) Development of student, recruiter, and administrator modules.
5) Integration of application, notification, resume-storage, and database workflows.
6) Functional and system-level testing.
7) Evaluation of usability and system performance.

B. Functional Evaluation

Functional evaluation was performed using automated regression tests covering authentication, authorization, ownership protections, resume processing, resume storage, Resume AI, Resume Builder, Job Match, Skill Gap, Courses, jobs, applications, recruiter workflows, administrator operations, and security controls.

The final hardening validation recorded **287 passed tests, 5 intentionally skipped tests, and 0 failed tests**. The intentionally skipped tests require a dedicated PostgreSQL/cutover environment and are not treated as failures.

The functional success rate is expressed as:

$$\text{Success Rate} = \frac{N_{\text{passed}}}{N_{\text{executed}}} \times 100 \tag{2}$$

where N_{passed} is the number of successfully completed tests and N_{executed} is the number of executed tests.

C. Performance Evaluation

Performance work focused particularly on Job Match because semantic matching can become computationally expensive as the number of jobs increases. The implementation uses a singleton semantic model, reuses the resume embedding during a matching operation, paginates ordinary job retrieval, and bounds Job Match evaluation to the 100 newest active jobs.

These values describe the tested local deterministic fallback path and should not be interpreted as production-scale throughput measurements. Sentence Transformer cold-start timing was not treated as a reliable benchmark because model availability and local caching can affect initialization time.

TABLE I
JOB MATCH DETERMINISTIC FALLBACK BENCHMARK

Condition	Jobs	Time (s)
Cold fallback	10	4.091
Warm fallback	50	0.748
Warm fallback	100	1.542

D. Usability Evaluation

A formal user study with a statistically representative participant population has not been completed. Therefore, the study does not report an unsupported numerical usability score. Usability is considered through task-oriented workflows, interface consistency, validation feedback, and error handling. A controlled user study remains an appropriate future evaluation.

VII. RESULTS AND DISCUSSION

The current implementation demonstrates an integrated workflow spanning resume preparation, intelligent resume analysis, job matching, skill-gap identification, learning support, application management, recruiter operations, and administrative placement management.

A. Functional Results

The final automated regression suite recorded:

287 passed 5 skipped 0 failed

The tested functionality includes:

- role-based authentication and authorization;
- CSRF protection for state-changing operations;
- resume upload, replacement, parsing, OCR fallback, and deletion;
- private resume storage and ownership protection;
- canonical Resume AI scoring;
- Resume Builder template handling;
- Job Match ranking and fallback behavior;
- Jobs pagination and candidate bounding;
- Skill Gap and Courses workflows;
- recruiter company and applicant workflows;
- rejection feedback and application status handling; and
- administrative management operations.

The absence of test failures in the final hardening suite provides evidence that the implemented regression scenarios behaved as expected under the tested environment.

B. Resume Intelligence Results

The implemented resume pipeline supports both normal text extraction and OCR fallback. The structured representation is reused by Resume AI and Job Match rather than independently reinterpreting the same document in different modules. This establishes a canonical flow from the resume file to structured analysis, Resume AI metrics, Job Match, and Skill Gap analysis.

C. Job Matching Results

The hybrid Job Match design combines deterministic requirement checks with semantic similarity. This provides complementary signals from explicit job requirements and contextual similarity between the resume and job description. Canonical skill handling reduces common false-positive matches, such as treating Java and JavaScript as identical or treating SQL, MySQL, and PostgreSQL as interchangeable credentials.

D. Database and Storage Results

PostgreSQL/Supabase integration was validated together with private resume storage. The application maintains resume metadata in PostgreSQL while storing files in a private Supabase Storage bucket. Ownership and access-control tests verify that users cannot access resources belonging to unrelated accounts through protected workflows.

E. Performance Results

The application includes database-level job pagination and bounded Job Match candidate evaluation. Model reuse and one-time resume embedding computation reduce repeated initialization and embedding costs during a matching operation. The deterministic fallback measurements in Table I show sub-two-second warm matching for the tested 50- and 100-job cases. These results are environment-specific and are presented as implementation benchmarks rather than generalized production guarantees.

F. Discussion

The results indicate that CareerCompass can integrate career preparation and institutional recruitment workflows without treating resume analysis, job matching, learning support, and application management as separate systems. The strongest architectural characteristic is the reuse of a structured active resume across multiple downstream features.

The evaluation also has limitations. A formal usability study has not been completed, and a completely fresh environment installation involving the full machine-learning dependency stack was not fully verified because the large PyTorch dependency installation exceeded the available validation window. These limitations are reported explicitly rather than converted into unsupported performance or usability claims.

VIII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

The current study has several limitations. First, formal usability evaluation with representative participants has not yet been completed. Second, the reported Job Match benchmark is based on a local deterministic fallback and a bounded candidate set; comprehensive production-scale benchmarking with a larger dedicated PostgreSQL fixture database remains future work. Third, the full clean-environment installation of the machine-learning dependency stack was initiated but not completed within the available validation window. Finally, current Job Match evaluation is bounded to the newest active jobs, which provides a predictable computational limit but can exclude older jobs from a matching request.

B. Future Work

Future development can build on the current platform through:

- **Job Trust Engine:** combine recruiter/company verification, job freshness, duplicate detection, suspicious-listing indicators, application-destination checks, and user reports to provide evidence-based job-trust information.
- **Application Intelligence:** provide pre-application explanations of resume readiness, missing requirements, and job-specific preparation opportunities.
- **Adaptive Career Roadmap:** connect a target role with skill gaps, learning resources, projects, resume improvements, and subsequent re-evaluation.
- **Application Outcome Intelligence:** analyze views, shortlisting, interviews, rejection, and offers to help users understand application performance.
- **Advanced Job Matching:** incorporate additional signals such as work mode, location preferences, salary requirements, job freshness, and verified-job indicators.
- **Interview Preparation:** provide job-specific preparation material and structured mock-interview workflows.
- **External Integrations:** connect with external job and learning platforms where appropriate.
- **Advanced Analytics:** provide aggregated placement and recruitment analytics for authorized institutional users.
- **Mobile and multilingual access:** extend the platform to mobile clients and additional languages.
- **AI Career Assistant:** provide conversational career support grounded in the student's verified profile, resume, skills, and application history.

IX. Conclusion

CareerCompass is an integrated web-based career guidance and placement management platform that connects student career preparation with recruiter and administrative placement workflows. Unlike a conventional placement portal that primarily manages profiles, jobs, and applications, the current implementation connects an active resume with downstream career-support functions including Resume AI, Job Match, Skill Gap analysis, and Courses.

The platform uses Flask, PostgreSQL, and Supabase services, including private cloud storage for resumes. Its resume-processing pipeline supports validated PDF upload, structured extraction, OCR fallback, canonical analysis, and controlled storage. The Job Match engine combines explicit requirement evaluation with local semantic similarity and a deterministic fallback, while performance safeguards reduce repeated model and embedding computation.

The final automated hardening validation recorded 287 passed tests, 5 intentionally skipped tests, and no failures. These results provide evidence of functional correctness across the tested workflows, including security, resume, job, application, recruiter, administrator, and intelligent career-support components. Formal usability evaluation and broader production-scale performance testing remain areas for future work.

CareerCompass therefore demonstrates the feasibility of combining career preparation, intelligent resume-to-job analysis, learning support, and institutional recruitment management within a single platform. Future work can extend this foundation toward job-trust analysis, application intelligence, adaptive career planning, interview preparation, and outcome-driven career support.

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